# Answer Key & Teacher Notes

Session 1 — Multiplication Equations · Session 2 — Division Equations

## HOW TO USE THIS LESSON'S PAGES

*one lesson = two sessions*

| | |
|---|---|
| **Warm-Up** | About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key. |
| **Prerequisite Practice A/B** | Before the lesson, for students who need the skill the session uses. Key in that folder. |
| **Guided Notes** | During the lesson. Students fill in the blanks as you teach; answers below. |
| **Practice Form A** | In class, after the lesson. Two pages. The self-check code tells students if they are on track. |
| **Practice Form B** | Same skills, new numbers: reteach, retake, small group or absent students. |
| **Exit Ticket** | Last 5 minutes. Two per page; cut in half. |
| **Homework Packet** | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 · TEACHER NOTES

*Multiplication Equations*

### TALK PROMPTS

- Why does dividing both sides by the coefficient leave the variable by itself?
- How do you know the 7 is the coefficient and not the solution?

### PARTNER TASK

Form A, problems 3 and 4. One partner writes and solves Bookshelf Rows (7b = 133); the other checks by multiplying. Switch roles with Egg Cartons (15e = 270).

### WATCH FOR

- Multiplying by the coefficient instead of dividing (5x = 65 gives 325).
- Subtracting the coefficient (5x = 65 gives 60).
- Dividing the coefficient by the total (5 ÷ 65) instead of the total by the coefficient.

## SESSION 2 · TEACHER NOTES

*Division Equations*

### TALK PROMPTS

- Why does multiplying both sides by the divisor get the variable alone?
- How does the total compare with the size of one share, and why does that make sense?

### PARTNER TASK

Costume Sashes. A spool of ribbon is cut evenly into 9 sashes of 16 inches. One partner writes and solves n ÷ 9 = 16; the other checks by dividing. Switch roles with Class Pizza (s ÷ 5 = 12).

### WATCH FOR

- Dividing again instead of multiplying (h ÷ 5 = 20 gives 4).
- Accepting a total that is smaller than one share.
- Using the wrong number of groups (homework: Kayla and 3 friends is 4 people, not 3).

## ITEM ANALYSIS

*suggested DOK; judge the reasoning required, not the question label*

| Sheet | Standard | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **S1 · Form A** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |
| **S1 · Form B** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |
| **S2 · Form A** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 2 |
| **S2 · Form B** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |

*Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.*

### GUIDED NOTES

**coefficient:** The number that **multiplies** the variable.

**inverse operations:** × and ÷ **undo** each other.

**Division Property of Equality:** Divide both sides by the same number (not **zero**). They stay equal.

- Division **undoes** multiplication.
- To solve $5b = 45$, divide both sides by the **coefficient**, 5.

**Example:** The coefficient is **6**. Divide both sides by 6: $c = 84 \div 6 =$ **14**. Check: $6 \times 14 =$ **84**. Answer: $c =$ **14** chairs in each stack

### YOU TRY

**1.** a. x = 9 b. n = 13

**2.** w = 3.2 ÷ 4 = 0.8. Check: 4 × 0.8 = 3.2.

### EXIT TICKET

**1.** a. x = 8 b. k = 1.5

**2.** 8t = 96; t = $12

**3.** **B** (coefficient)

### PRACTICE FORM A

**1.** a. k = 5 b. m = 13 c. w = 8 d. y = 2.5 e. t = 6 f. r = 3.5

**2.** a. 5p = 175 b. p = 175 ÷ 5 = $35 c. 5 × 35 = 175

**3.** 7b = 133; b = 19 books

**4.** e = 270 ÷ 15 = 18 eggs

**5.** m = 154 ÷ 14 = 11 miles

**6.** 24w = 336; w = $14

**7.** 18c = 216; c = 12 cups. Check: 18 × 12 = 216.

**8.** a. 24r = 648; r = 27 rows b. 17 × 24 = 408 seats c. Multiply: 24 × 27 = 648, so the answer checks.

**9.** Multiplying does not undo × 5. She should divide both sides by 5: x = 65 ÷ 5 = 13. Check: 5 × 13 = 65.

**10.** **C** ($n = 12$)

**11.** **B** (coefficient)

**Self-check word** **PRODUCT**

### PRACTICE FORM B

**1.** a. k = 6 b. m = 17 c. w = 7 d. y = 3.5 e. t = 4 f. r = 4.5

**2.** a. 6p = 258 b. p = 258 ÷ 6 = $43 c. 6 × 43 = 258

**3.** 8b = 168; b = 21 books

**4.** e = 264 ÷ 12 = 22 eggs

**5.** m = 208 ÷ 16 = 13 miles

**6.** 28w = 420; w = $15

**7.** 16c = 224; c = 14 cups. Check: 16 × 14 = 224.

**8.** a. 26r = 754; r = 29 rows b. 19 × 26 = 494 seats c. Multiply: 26 × 29 = 754, so the answer checks.

**9.** Multiplying does not undo × 6. He should divide both sides by 6: x = 78 ÷ 6 = 13. Check: 6 × 13 = 78.

**10.** **B** ($n = 12$)

**11.** **C** (divide both sides by the coefficient)

**Self-check word** **DIVIDES**

## SESSION 2 · ANSWERS

### GUIDED NOTES

**divisor:** The number you **divide** by.

**Multiplication Property of Equality:** Multiply **both** sides by the same number. They stay equal.

**equal groups:** Groups that each have the **same** amount.

- Multiplication **undoes** division.
- To solve $n \div 4 = 7$, multiply both sides by **4**.

**Example:** Multiply both sides by **9**. $t = 7 \times 9 = \mathbf{63}$. Check: $63 \div 9 = \mathbf{7}$. Answer: The bag holds **63** ounces.

### YOU TRY

**1.** a. $x = 36$ b. $n = 56$

**2.** $n = 1.5 \times 2 = 3$. Check: $3 \div 2 = 1.5$.

### EXIT TICKET

**1.** a. $n = 48$ b. $k = 10$

**2.** $r \div 5 = 3.5$; $r = 17.5$ m

**3.** **B** (multiply both sides by 5)

### PRACTICE FORM A

**1.** a. $k = 63$ b. $p = 60$ c. $m = 96$ d. $n = 78$ e. $y = 9$ f. $w = 36$

**2.** a. $s \div 5 = 12$ b. $s = 12 \times 5 = 60$ slices c. $60 \div 5 = 12$

**3.** $g \div 4 = 23.50$; $g = \$94$

**4.** $n = 16 \times 9 = 144$ inches

**5.** $s = 14 \times 15 = 210$ seeds

**6.** $w \div 12 = 16$; $w = 192$ liters

**7.** $p \div 7 = 24$; $p = 168$ pages

**8.** a. $t \div 14 = 45$; $t = 630$ tiles b. $18 \times 45 = 810$ tiles c. Divide: $630 \div 14 = 45$, so it checks.

**9.** Dividing again does not undo ÷ 5. He should multiply both sides by 5: $h = 20 \times 5 = 100$. Check: $100 \div 5 = 20$.

**10.** **A** ($x = 55$)

**11.** **C** (greater than 20)

**Self-check word** **DIVISOR**

### PRACTICE FORM B

**1.** a. $k = 72$ b. $p = 70$ c. $m = 66$ d. $n = 84$ e. $y = 10$ f. $w = 48$

**2.** a. $s \div 6 = 11$ b. $s = 11 \times 6 = 66$ slices c. $66 \div 6 = 11$

**3.** $g \div 3 = 31.50$; $g = \$94.50$

**4.** $n = 18 \times 7 = 126$ inches

**5.** $s = 15 \times 13 = 195$ seeds

**6.** $w \div 14 = 15$; $w = 210$ liters

**7.** $p \div 6 = 26$; $p = 156$ pages

**8.** a. $t \div 16 = 35$; $t = 560$ tiles b. $20 \times 35 = 700$ tiles c. Divide: $560 \div 16 = 35$, so it checks.

**9.** Dividing again does not undo ÷ 6. She should multiply both sides by 6: $m = 12 \times 6 = 72$. Check: $72 \div 6 = 12$.

**10.** **C** ($n = 72$)

**11.** **B** (multiply both sides by the divisor)

**Self-check word** **PRODUCT**

## HOMEWORK PACKET · ANSWERS

*same answers for the core packet and Level 1*

### SHEET 1 · MULTIPLICATION EQUATIONS

**1.** a. $x = 9$ b. $n = 8$ c. $y = 12$ d. $k = 7$ e. $m = 17$ f. $w = 1.5$

**2.** $5c = 65$; $c = 13$

**3.** $5f = 11.25$; $f = \$2.25$. Check: $5 \times 2.25 = 11.25$.

**4.** $6s = 84$; $s = 14$ seedlings. Check: $6 \times 14 = 84$.

**5.** $4n = 80$ has the greater solution. The same total split into fewer equal groups makes each group larger. Check: $4n = 80$ gives $n = 20$; $5n = 80$ gives $n = 16$.

### SHEET 2 · DIVISION EQUATIONS

**1.** a. $n = 42$ b. $y = 65$ c. $k = 45$

**2.** $t \div 3 = 17$; $t = 51$

**3.** $r \div 8 = 2.5$; $r = 20$ meters

**4.** $z \div 6 = 2$; $z = 12$ slices. Check: $12 \div 6 = 2$.

**5.** The cards are shared by 4 people (Kayla and 3 friends), not 3. Correct equation: $c \div 4 = 9$, so $c = 36$ cards.

**6.** **D** ($y = 54$)

**7.** **A** ($p \div 12 = 5$)

# October 2026 conceptual additions — teacher key

## Session 1 · Multiplication Equations

6g=42; g=7; six equal groups of seven total 42.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: 6g=42; g=7; six equal groups of seven total 42.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

## Session 2 · Division Equations

t/5=9; multiply both sides by 5: t=45. Five positive equal shares make the total exceed one share; this is a contextual condition, not a universal division rule.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: t/5=9; multiply both sides by 5: t=45. Five positive equal shares make the total exceed one share; this is a contextual condition, not a universal division rule.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.